

Performance metrics across experimental scenarios: all decoders

PLOS ONE table-layout preview

Generated from the completed scenario_decoder sweep (2026-08-24 — 15-13-34).

Logistic regression

Table 1. Performance metrics across experimental scenarios for logistic regression.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.64	0.64	0.28	0.71	0.62/0.66
	Stratified K-fold(Full B)	0.68	0.65	0.33	0.73	0.65/0.71
	Stratified K-fold(B: Stroop)	0.69	0.69	0.39	0.74	0.67/0.71
	Stratified K-fold(B: Arithmetic)	0.66	0.66	0.33	0.72	0.66/0.67
	Stratified K-fold(B: Mirror)	0.69	0.69	0.37	0.73	0.66/0.72
Cross-subject	Leave-one-subject-out(Full A)	0.63	0.63	0.27	0.69	0.63/0.64
	Leave-one-subject-out(Full B)	0.66	0.63	0.29	0.71	0.63/0.69
Cross-session	Leave-one-session-out(Full A)	0.63	0.63	0.26	0.67	0.55/0.70
Cross-task (Dataset B)	Stroop → Arithmetic	0.59	0.59	0.19	0.63	0.65/0.54
	Stroop → Mirror	0.66	0.66	0.33	0.70	0.67/0.66
	Arithmetic → Stroop	0.56	0.56	0.12	0.60	0.67/0.46
	Arithmetic → Mirror	0.61	0.58	0.20	0.65	0.65/0.56
	Mirror → Stroop	0.62	0.62	0.25	0.68	0.63/0.62
	Mirror → Arithmetic	0.60	0.60	0.19	0.63	0.66/0.54
	Arithmetic+Mirror → Stroop	0.61	0.58	0.20	0.65	0.65/0.56
	Stroop+Mirror → Arithmetic	0.66	0.66	0.33	0.72	0.59/0.74
	Stroop+Arithmetic → Mirror	0.62	0.62	0.25	0.67	0.62/0.63
	Stroop → Arithmetic+Mirror	0.58	0.57	0.16	0.60	0.65/0.50
	Arithmetic → Stroop+Mirror	0.60	0.58	0.20	0.63	0.63/0.58
	Mirror → Stroop+Arithmetic	0.63	0.63	0.25	0.66	0.64/0.62
	Full A → Full B	0.52	0.39	0.04	0.54	0.78/0.26
	Full B → Full A	0.55	0.53	0.12	0.58	0.35/0.76
	Full A → Stroop	0.49	0.46	-0.03	0.50	0.71/0.27
Cross-dataset	Stroop → Full A	0.54	0.54	0.08	0.56	0.45/0.63
	Full A → Arithmetic	0.53	0.52	0.07	0.56	0.68/0.39
	Arithmetic → Full A	0.53	0.52	0.05	0.54	0.44/0.61
	Full A → Mirror	0.53	0.51	0.07	0.54	0.72/0.34
	Mirror → Full A	0.53	0.49	0.06	0.55	0.26/0.80
	Full A → Stroop+Arithmetic	0.51	0.44	0.02	0.53	0.72/0.30
	Stroop+Arithmetic → Full A	0.52	0.51	0.05	0.54	0.36/0.68
	Full A → Stroop+Mirror	0.51	0.43	0.02	0.52	0.75/0.27
	Stroop+Mirror → Full A	0.53	0.50	0.06	0.55	0.28/0.78
	Full A → Arithmetic+Mirror	0.54	0.46	0.08	0.56	0.77/0.30
	Arithmetic+Mirror → Full A	0.55	0.51	0.11	0.59	0.28/0.81

Note. Per-class recall is reported as Low/High. MCC, Matthews correlation coefficient; ROC-AUC, area under the receiver operating characteristic curve. Dashes indicate values to be filled.

XGBoost

Table 2. Performance metrics across experimental scenarios for XGBoost.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.67	0.67	0.34	0.73	0.64/0.70
	Stratified K-fold(Full B)	0.68	0.66	0.33	0.74	0.60/0.76
	Stratified K-fold(B: Stroop)	0.69	0.69	0.38	0.76	0.68/0.70
	Stratified K-fold(B: Arithmetic)	0.70	0.69	0.39	0.75	0.69/0.70
	Stratified K-fold(B: Mirror)	0.70	0.70	0.41	0.76	0.72/0.69
Cross-subject	Leave-one-subject-out(Full A)	0.66	0.66	0.32	0.72	0.64/0.67
	Leave-one-subject-out(Full B)	0.65	0.63	0.28	0.71	0.58/0.72
Cross-session	Leave-one-session-out(Full A)	0.65	0.65	0.31	0.70	0.53/0.77
Cross-task (Dataset B)	Stroop → Arithmetic	0.61	0.61	0.22	0.64	0.65/0.58
	Stroop → Mirror	0.67	0.67	0.33	0.72	0.66/0.67
	Arithmetic → Stroop	0.58	0.58	0.17	0.62	0.68/0.48
	Arithmetic → Mirror	0.62	0.60	0.23	0.67	0.67/0.58
	Mirror → Stroop	0.64	0.64	0.29	0.69	0.70/0.59
	Mirror → Arithmetic	0.61	0.61	0.24	0.65	0.72/0.51
	Arithmetic+Mirror → Stroop	0.62	0.59	0.23	0.67	0.70/0.55
	Stroop+Mirror → Arithmetic	0.66	0.66	0.33	0.72	0.60/0.72
	Stroop+Arithmetic → Mirror	0.61	0.61	0.22	0.66	0.63/0.59
	Stroop → Arithmetic+Mirror	0.56	0.55	0.12	0.58	0.67/0.45
	Arithmetic → Stroop+Mirror	0.58	0.56	0.16	0.62	0.63/0.53
	Mirror → Stroop+Arithmetic	0.62	0.62	0.24	0.66	0.65/0.59
	Full A → Full B	0.52	0.52	0.03	0.52	0.28/0.75
	Full B → Full A	0.54	0.53	0.09	0.55	0.37/0.71
	Full A → Stroop	0.48	0.45	-0.05	0.47	0.26/0.70
Cross-dataset	Stroop → Full A	0.51	0.49	0.02	0.51	0.33/0.69
	Full A → Arithmetic	0.54	0.50	0.10	0.55	0.27/0.81
	Arithmetic → Full A	0.59	0.59	0.18	0.62	0.62/0.56
	Full A → Mirror	0.51	0.48	0.03	0.53	0.27/0.76
	Mirror → Full A	0.53	0.53	0.07	0.55	0.61/0.46
	Full A → Stroop+Arithmetic	0.52	0.52	0.04	0.51	0.29/0.75
	Stroop+Arithmetic → Full A	0.53	0.52	0.06	0.53	0.37/0.68
	Full A → Stroop+Mirror	0.50	0.50	0.00	0.50	0.27/0.73
	Stroop+Mirror → Full A	0.52	0.51	0.03	0.52	0.38/0.66
	Full A → Arithmetic+Mirror	0.52	0.52	0.05	0.54	0.27/0.78
	Arithmetic+Mirror → Full A	0.56	0.56	0.13	0.58	0.57/0.56

Note. Per-class recall is reported as Low/High. MCC, Matthews correlation coefficient; ROC-AUC, area under the receiver operating characteristic curve. Dashes indicate values to be filled.

EEGNet

Table 3. Performance metrics across experimental scenarios for EEGNet.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.66	0.66	0.33	0.74	0.57/0.76
	Stratified K-fold(Full B)	0.67	0.66	0.32	0.73	0.57/0.77
	Stratified K-fold(B: Stroop)	0.66	0.66	0.33	0.71	0.58/0.74
	Stratified K-fold(B: Arithmetic)	0.65	0.64	0.29	0.70	0.60/0.69
	Stratified K-fold(B: Mirror)	0.68	0.67	0.36	0.73	0.60/0.76
Cross-subject	Leave-one-subject-out(Full A)	0.66	0.66	0.31	0.72	0.65/0.66
	Leave-one-subject-out(Full B)	0.67	0.64	0.30	0.71	0.60/0.74
Cross-session	Leave-one-session-out(Full A)	0.63	0.63	0.27	0.67	0.59/0.68
Cross-task (Dataset B)	Stroop → Arithmetic	0.62	0.62	0.25	0.65	0.59/0.66
	Stroop → Mirror	0.65	0.65	0.31	0.69	0.60/0.71
	Arithmetic → Stroop	0.59	0.59	0.18	0.62	0.61/0.57
	Arithmetic → Mirror	0.62	0.61	0.24	0.66	0.61/0.64
	Mirror → Stroop	0.62	0.62	0.25	0.66	0.57/0.68
	Mirror → Arithmetic	0.61	0.61	0.23	0.66	0.57/0.65
	Arithmetic+Mirror → Stroop	0.61	0.61	0.22	0.66	0.57/0.66
	Stroop+Mirror → Arithmetic	0.67	0.67	0.34	0.72	0.61/0.73
	Stroop+Arithmetic → Mirror	0.61	0.61	0.23	0.66	0.60/0.63
	Stroop → Arithmetic+Mirror	0.56	0.55	0.12	0.58	0.65/0.47
	Arithmetic → Stroop+Mirror	0.58	0.56	0.14	0.60	0.59/0.56
	Mirror → Stroop+Arithmetic	0.59	0.59	0.19	0.64	0.60/0.58
	Full A → Full B	0.47	0.39	-0.06	0.47	0.61/0.32
	Full B → Full A	0.51	0.49	0.03	0.53	0.30/0.73
	Full A → Stroop	0.45	0.43	-0.11	0.43	0.60/0.29
Cross-dataset	Stroop → Full A	0.51	0.48	0.01	0.51	0.27/0.74
	Full A → Arithmetic	0.49	0.49	-0.02	0.51	0.58/0.40
	Arithmetic → Full A	0.50	0.44	0.01	0.52	0.16/0.85
	Full A → Mirror	0.46	0.45	-0.09	0.47	0.60/0.32
	Mirror → Full A	0.49	0.49	-0.02	0.48	0.44/0.54
	Full A → Stroop+Arithmetic	0.46	0.42	-0.07	0.47	0.60/0.32
	Stroop+Arithmetic → Full A	0.51	0.40	0.03	0.58	0.08/0.94
	Full A → Stroop+Mirror	0.45	0.40	-0.11	0.45	0.60/0.29
	Stroop+Mirror → Full A	0.50	0.39	0.01	0.50	0.08/0.93
	Full A → Arithmetic+Mirror	0.48	0.44	-0.03	0.49	0.60/0.37
	Arithmetic+Mirror → Full A	0.51	0.49	0.03	0.52	0.28/0.74

Note. Per-class recall is reported as Low/High. MCC, Matthews correlation coefficient; ROC-AUC, area under the receiver operating characteristic curve. Dashes indicate values to be filled.

ShallowNet

Table 4. Performance metrics across experimental scenarios for ShallowFBCSPNet.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.66	0.66	0.32	0.73	0.63/0.69
	Stratified K-fold(Full B)	0.72	0.72	0.44	0.81	0.58/0.86
	Stratified K-fold(B: Stroop)	0.78	0.78	0.57	0.86	0.77/0.79
	Stratified K-fold(B: Arithmetic)	0.75	0.75	0.50	0.83	0.76/0.74
	Stratified K-fold(B: Mirror)	0.73	0.73	0.46	0.80	0.72/0.74
Cross-subject	Leave-one-subject-out(Full A)	0.67	0.67	0.34	0.73	0.72/0.62
	Leave-one-subject-out(Full B)	0.66	0.66	0.32	0.75	0.52/0.81
Cross-session	Leave-one-session-out(Full A)	0.66	0.66	0.33	0.71	0.58/0.74
Cross-task (Dataset B)	Stroop → Arithmetic	0.63	0.63	0.26	0.67	0.63/0.63
	Stroop → Mirror	0.69	0.69	0.38	0.75	0.73/0.65
	Arithmetic → Stroop	0.62	0.62	0.25	0.68	0.72/0.53
	Arithmetic → Mirror	0.66	0.63	0.30	0.72	0.73/0.59
	Mirror → Stroop	0.68	0.68	0.36	0.74	0.69/0.68
	Mirror → Arithmetic	0.64	0.64	0.28	0.68	0.72/0.56
	Arithmetic+Mirror → Stroop	0.66	0.64	0.30	0.72	0.70/0.62
	Stroop+Mirror → Arithmetic	0.67	0.66	0.33	0.74	0.61/0.72
	Stroop+Arithmetic → Mirror	0.67	0.67	0.34	0.72	0.62/0.72
	Stroop → Arithmetic+Mirror	0.60	0.59	0.20	0.65	0.72/0.48
	Arithmetic → Stroop+Mirror	0.62	0.59	0.23	0.68	0.70/0.54
	Mirror → Stroop+Arithmetic	0.64	0.64	0.28	0.70	0.73/0.55
Cross-dataset	Full A → Full B	0.49	0.34	-0.03	0.48	0.79/0.19
	Full B → Full A	0.55	0.51	0.11	0.56	0.28/0.81
	Full A → Stroop	0.47	0.41	-0.09	0.43	0.77/0.16
	Stroop → Full A	0.55	0.55	0.11	0.58	0.52/0.59
	Full A → Arithmetic	0.53	0.51	0.07	0.55	0.74/0.32
	Arithmetic → Full A	0.56	0.56	0.11	0.59	0.55/0.56
	Full A → Mirror	0.48	0.43	-0.05	0.48	0.78/0.19
	Mirror → Full A	0.55	0.55	0.11	0.58	0.52/0.58
	Full A → Stroop+Arithmetic	0.51	0.40	0.01	0.49	0.80/0.21
	Stroop+Arithmetic → Full A	0.53	0.52	0.06	0.55	0.37/0.69
	Full A → Stroop+Mirror	0.48	0.36	-0.06	0.45	0.78/0.17
	Stroop+Mirror → Full A	0.57	0.57	0.14	0.58	0.59/0.55
	Full A → Arithmetic+Mirror	0.51	0.41	0.02	0.52	0.78/0.24
	Arithmetic+Mirror → Full A	0.55	0.54	0.10	0.58	0.64/0.45

Note. Per-class recall is reported as Low/High. MCC, Matthews correlation coefficient; ROC-AUC, area under the receiver operating characteristic curve. Dashes indicate values to be filled.

EEGConformer

Table 5. Performance metrics across experimental scenarios for EEGConformer.

Scenario	Validation method	Balanced Accuracy	Macro F1	MCC	ROC AUC	Per-class Recall (Low/High)
Baseline (within)	Stratified K-fold(Full A)	0.65	0.65	0.30	0.72	0.69/0.60
	Stratified K-fold(Full B)	0.70	0.70	0.39	0.78	0.56/0.84
	Stratified K-fold(B: Stroop)	0.72	0.71	0.44	0.79	0.59/0.84
	Stratified K-fold(B: Arithmetic)	0.72	0.72	0.45	0.79	0.61/0.83
	Stratified K-fold(B: Mirror)	0.69	0.68	0.38	0.77	0.59/0.78
Cross-subject	Leave-one-subject-out(Full A)	0.65	0.65	0.30	0.72	0.65/0.64
	Leave-one-subject-out(Full B)	0.71	0.70	0.40	0.75	0.60/0.82
Cross-session	Leave-one-session-out(Full A)	0.66	0.66	0.32	0.70	0.58/0.74
Cross-task (Dataset B)	Stroop → Arithmetic	0.62	0.62	0.24	0.66	0.65/0.59
	Stroop → Mirror	0.68	0.68	0.36	0.73	0.62/0.73
	Arithmetic → Stroop	0.60	0.60	0.20	0.64	0.61/0.59
	Arithmetic → Mirror	0.64	0.62	0.26	0.69	0.62/0.66
	Mirror → Stroop	0.67	0.67	0.35	0.72	0.64/0.71
	Mirror → Arithmetic	0.62	0.62	0.25	0.64	0.65/0.60
	Arithmetic+Mirror → Stroop	0.65	0.63	0.28	0.68	0.65/0.65
	Stroop+Mirror → Arithmetic	0.69	0.68	0.39	0.76	0.56/0.81
	Stroop+Arithmetic → Mirror	0.65	0.65	0.30	0.69	0.61/0.69
	Stroop → Arithmetic+Mirror	0.55	0.55	0.11	0.58	0.67/0.44
	Arithmetic → Stroop+Mirror	0.59	0.57	0.17	0.62	0.62/0.56
	Mirror → Stroop+Arithmetic	0.63	0.63	0.26	0.67	0.63/0.63
	Full A → Full B	0.47	0.36	-0.06	0.46	0.70/0.24
	Full B → Full A	0.58	0.58	0.17	0.62	0.62/0.55
	Full A → Stroop	0.42	0.39	-0.18	0.40	0.65/0.20
Cross-dataset	Stroop → Full A	0.52	0.44	0.07	0.61	0.14/0.91
	Full A → Arithmetic	0.52	0.52	0.05	0.54	0.60/0.45
	Arithmetic → Full A	0.60	0.59	0.20	0.63	0.68/0.51
	Full A → Mirror	0.47	0.45	-0.06	0.46	0.66/0.29
	Mirror → Full A	0.54	0.51	0.08	0.57	0.76/0.32
	Full A → Stroop+Arithmetic	0.48	0.41	-0.04	0.46	0.68/0.27
	Stroop+Arithmetic → Full A	0.58	0.58	0.17	0.62	0.58/0.58
	Full A → Stroop+Mirror	0.44	0.37	-0.13	0.43	0.65/0.23
	Stroop+Mirror → Full A	0.55	0.52	0.11	0.59	0.31/0.78
	Full A → Arithmetic+Mirror	0.50	0.44	0.00	0.50	0.68/0.32
	Arithmetic+Mirror → Full A	0.59	0.59	0.18	0.62	0.60/0.58

Note. Per-class recall is reported as Low/High. MCC, Matthews correlation coefficient; ROC-AUC, area under the receiver operating characteristic curve. Dashes indicate values to be filled.